the analogy that built chemical engineering

Reynolds · Prandtl · Chilton–Colburn · Bird–Stewart–Lightfoot

circuit thinking

every *linear* steady transport problem is a resistor network — which is why finite-volume codes and bond graphs look the same

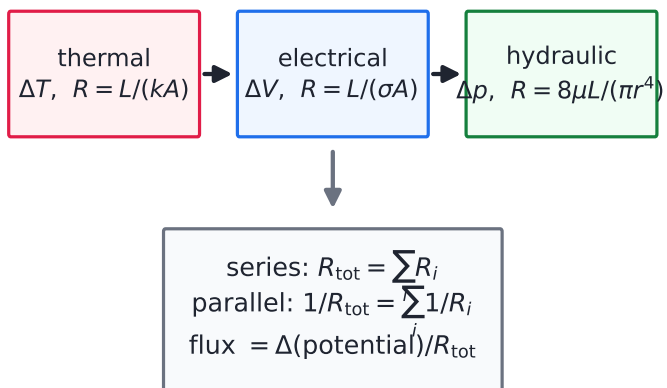
heat $q = -k\nabla T$ potential: T flux ratio: Nu

Fourier

mass $J = -D\nabla c$ potential: c flux ratio: Sh

Fick

momentum $\tau = \mu\nabla u$ potential: u flux ratio: C_f

Newton

flux $= -$(transport property) $\times \nabla$(potential)